Lingaro: a Revenue Growth Management pricing product, prototyped on real retail data

Price elasticity, promotion effectiveness and price-pack opportunities for an FMCG portfolio, packaged as a Power BI product with user stories and business rules: what I would deliver in Lingaro's RGM product team.

REAL RETAIL DATA · P.4

-2.9

price elasticity of Frozen Pizza: the most price-sensitive category (Bag Snacks -1.2)

REAL RETAIL DATA · P.5

+328%

volume uplift when feature and display run together, vs +33% for a price cut alone

REAL DATA + ASSUMED MARGIN · P.5

4/67

brand x mechanic combinations that pay back at an assumed 30% margin

MODEL · P.7

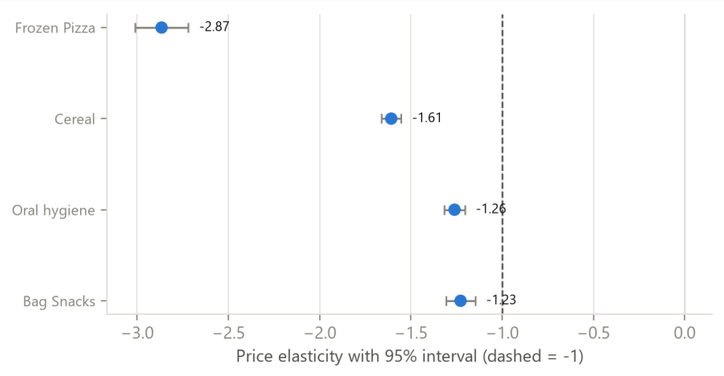
+12.3%

gross profit for +5% on General Mills cereal (volume -3.7%, range -3.9% to -3.6%)

Five questions an RGM product answers

- **Elasticity** - how price-sensitive is each category, brand and product? (p.4)
- **Promotions** - which mechanics create incremental volume, and which pay back? (p.5)
- **Price-pack** - is the price per ounce consistent across pack sizes? (p.6)
- **Scenarios** - what does +x% do to volume, revenue and profit? (p.7)
- **Scale** - does the same method hold market by market? (p.9)

Price elasticity by category (95% interval)



Recommendations

- Test small price increases first on the least price-sensitive brands (Shultz bag snacks, private-label bag snacks, General Mills cereal); hold price on frozen pizza, where an increase loses revenue.
- Move trade spend from TPR-only cuts (ROI -60% at 30% margin) toward feature + display events, and negotiate their fees.
- Fix the pack ladder before the next price list: Gm Cheerios 18 oz costs 4% more per ounce than Gm Honey Nut Cheerios 12.25 oz.
- Roll out market by market with the same rules; Indiana and Kentucky need more stores before their own elasticities can be published.

Data & methods

REAL RETAIL DATA dunnhumby *Breakfast at the Frat*: 524,950 store x product x weeks, 55 products, 77 US stores, 156 weeks (2009-2012).

PUBLIC DATA Lingaro website (dated facts), Eurostat HICP.

SIMULATED NielsenIQ / Circana-style market data, clearly labelled.

DATA QUALITY 99.6% of rows pass every check; 2 stores listed twice in the store file were de-duplicated; Colgate and Quaker leave the shelf in 2011 (kept in history, not simulated).

STACK SQL (DuckDB) · Python (statsmodels) · Power BI (TMDL + PBIR, 41 measures).

HOW TO READ Titles state the conclusion; the tag top right gives the data source; dashed lines are thresholds; ranges are 95% intervals.

Sources and licences: data/SOURCES.md · Assumptions: data/ASSUMPTIONS.md

Lingaro builds data & AI products for consumer goods; RGM pricing is one of them

PUBLIC DATA

Lingaro's public RGM case study describes exactly the product the RGM Data Analyst role supports: elasticity estimates and a what-if simulator, built once and scaled market by market.

Who Lingaro is

- Positions itself as a global end-to-end data and AI partner to leading brands and enterprises.
- On the market since 2008, 1,800+ people, 7 global sites: Warsaw, Lublin (PL); Geneva, Zurich (CH); Cincinnati (US); Mexico City (MX); Manila (PH); Bangalore (IN).
- Domains: Marketing & Commercial Analytics, Supply Chain, Sustainability & Compliance, AI & GenAI, Data Consulting.
- Named a Leader by ISG for retail & CPG analytics (2025); highest "value delivered" among Major Contenders in Everest Group's 2024 retail & CPG PEAK Matrix.
- Says it has partnered with 75% of Fortune 200 CPG companies (company statement).

Revenue Growth Management, in Lingaro's words

- RGM is listed as a Marketing & Commercial Analytics capability.
- Scope named on its site: price management, promo management, assortment, trade spend & profitability, **price elasticity analytics**, **what-if scenarios**, **promo effectiveness**, profit optimisation.
- On promotions: uplift can mix true incremental demand with stock-up, pull-forward, cannibalisation and seasonality; trade spend can be 20% to 30% of gross sales.
- Public case study: a price elasticity tool for a global beverage company - elasticity estimates, recommended strategies and what-if simulation on volume, sales value and gross profit, now the go-to source in 4 markets.

What this prototype mirrors

In Lingaro's case study / posting	Here
Elasticity estimates per product	p.4, log-log with 95% intervals
Separate real demand from stock-up and cannibalisation	p.5, baseline, dip, cannibalisation
What-if on volume, sales value, gross profit	p.7, simulator with ranges
Data limitations explained to users	BR-06 publication rule
Scaled to other markets	p.9, one template, many markets
User stories, acceptance criteria, business rules	p.10, `product/`
NielsenIQ / Circana data	p.8, simulated feed, labelled

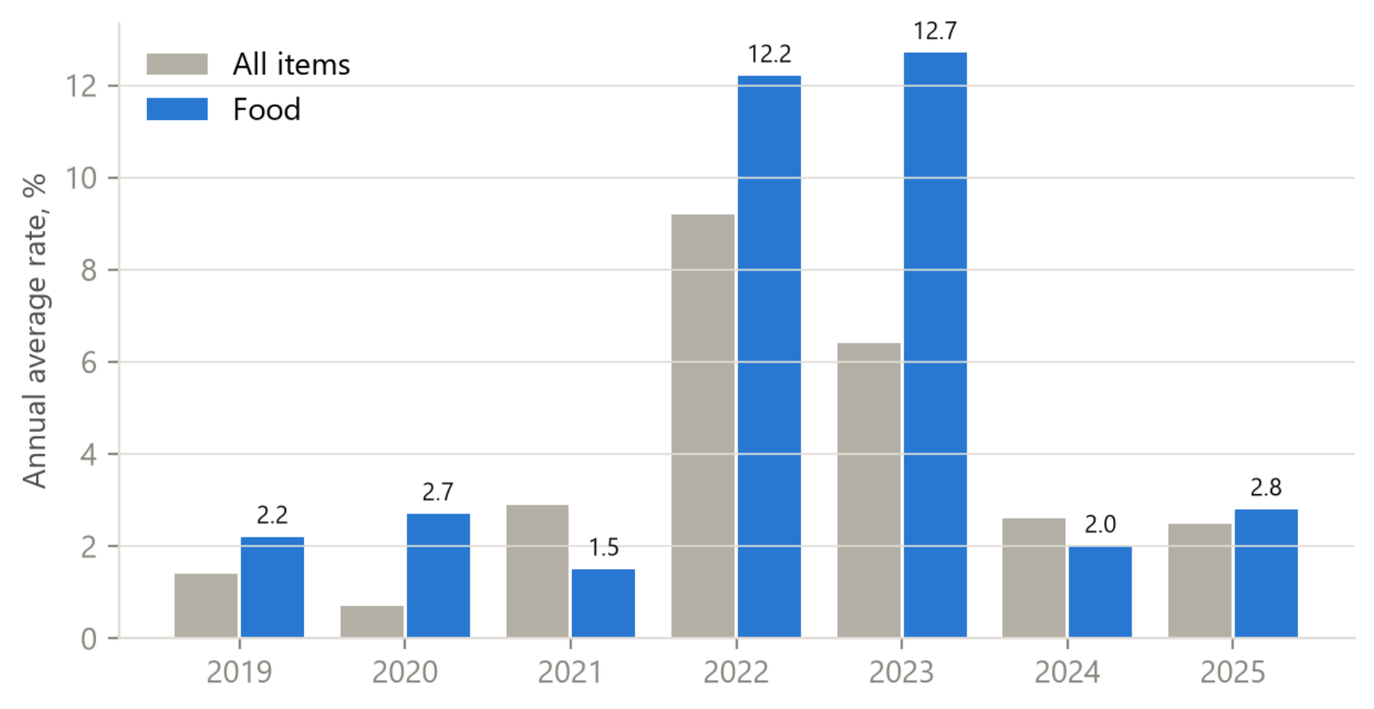
Source: lingarogroup.com (home, about, Marketing & Commercial Analytics, RGM case study, insights, careers), retrieved 2026-11-12 · data/public/lingaro_facts.csv

EU food prices rose 35% in five years: every price move now needs evidence

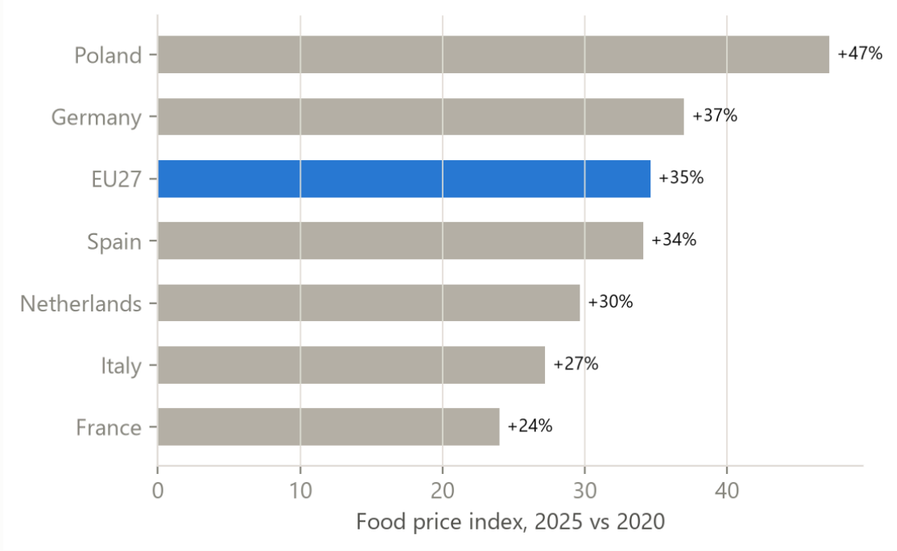
PUBLIC DATA

Food inflation in the EU peaked at 12.2% in 2022 and 12.7% in 2023, well above all items, before easing to 2.8% in 2025. Food prices outpaced the overall index (35% vs 26% since 2020).

EU27 HICP, annual average rate of change



Cumulative food price increase, 2020 to 2025



What it changes for revenue management

- After two years of double-digit food inflation, shoppers compare prices more; elasticity is no longer a constant to assume, it is a number to measure, per market.
- Retailers push back on list-price increases: each increase needs a volume, revenue and margin case.
- Promotions absorb more trade spend: knowing which ones pay back protects margin.

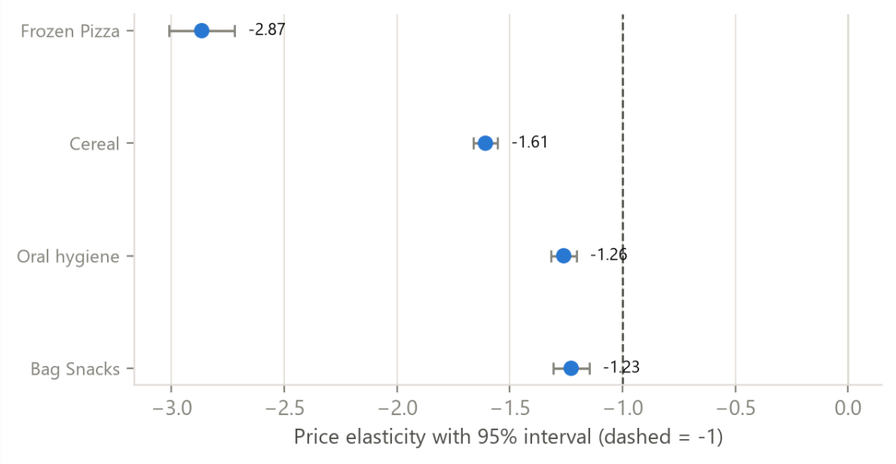
Source: Eurostat HICP prc_hicp_aind (CC BY 4.0), retrieved 2026-11-12

Frozen Pizza is the most price-sensitive category (-2.9); Bag Snacks the least (-1.2)

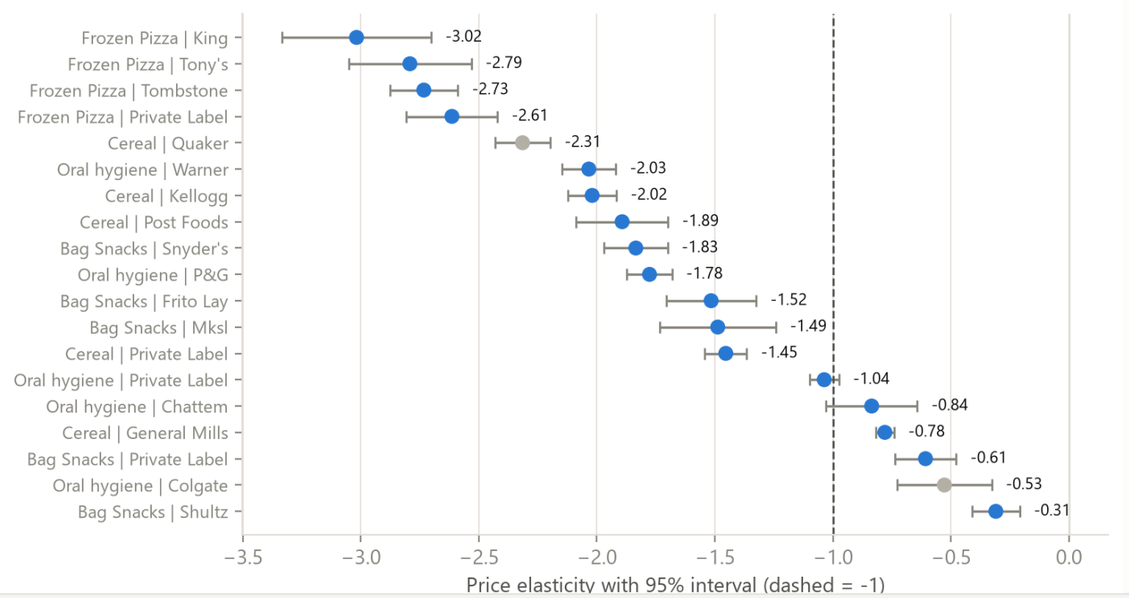
REAL RETAIL DATA

72 of 78 category, brand and product elasticities pass the publication rule (≥ 1,000 rows, ≥ 52 weeks, ≥ 5 stores, enough price variation, significant, negative, interval ≤ 1.5 wide); the other 6 are Colgate and Quaker, which leave the shelf in 2011. 11 of 19 brands show a significant substitution effect with the rest of their category.

By category (blue = published, BR-06)



By brand (manufacturer within category); grey = review



Reading "-2.9" in plain English

If Frozen Pizza prices go up 1%, units fall about 2.9%, all else equal. Because the fall is larger than 1%, revenue falls too. Bag Snacks (-1.2) is far less sensitive.

Model: log(units) on log(price), with feature, display, store x product and month effects and a trend; errors clustered by store. It is a **correlation on historical data**, not an experiment: prices were set by the retailer, sometimes alongside actions the data does not show.

Regular price vs promotion price cut

Category	Price component	Elasticity	CI low	CI high
Bag Snacks	Regular price	-1.61	-1.73	-1.49
Bag Snacks	Promo price cut	-1.00	-1.08	-0.93
Cold Cereal	Regular price	-1.70	-1.80	-1.59
Cold Cereal	Promo price cut	-1.58	-1.63	-1.53
Frozen Pizza	Regular price	-2.79	-2.93	-2.66
Frozen Pizza	Promo price cut	-2.90	-3.06	-2.74
Oral Hygiene Products	Regular price	-1.18	-1.25	-1.11
Oral Hygiene Products	Promo price cut	-1.28	-1.34	-1.23

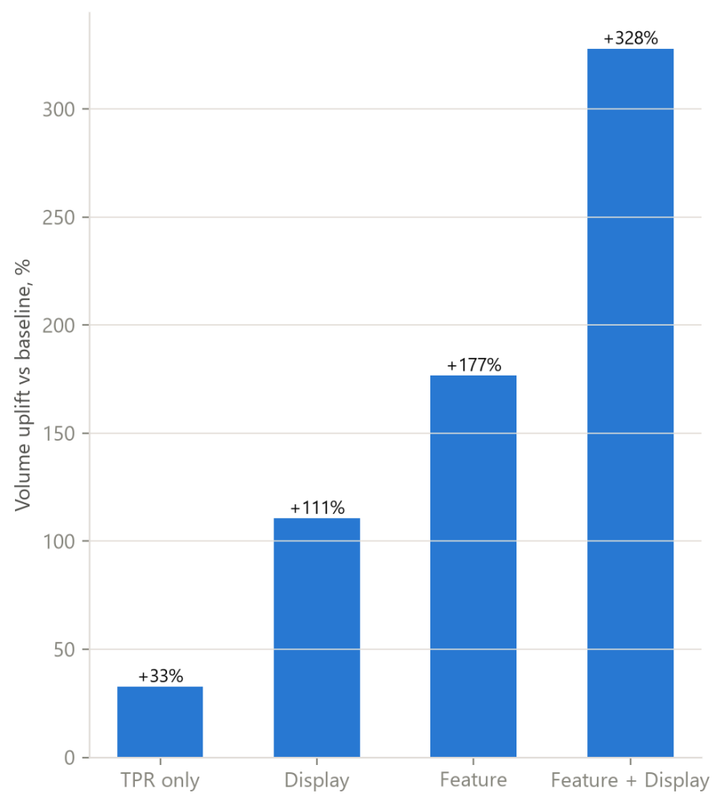
Source: dunnhumby 'Breakfast at the Frat' (licence: dunnhumby Source Files terms; raw data not redistributed) · src/elasticity.py

Feature + display lifts volume +328%; most price cuts do not pay back

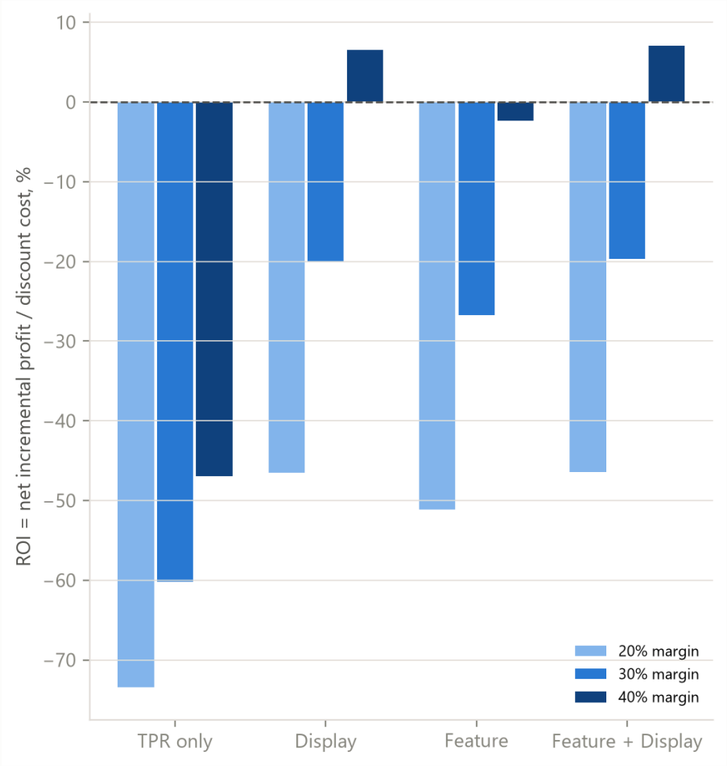
REAL RETAIL DATA · ASSUMED MARGIN

Uplift = promo-week units vs a baseline of clean non-promo weeks within ±8 weeks (BR-03). ROI = (profit with the promotion - profit without it) / discount cost, at an ASSUMED gross margin (BR-09).

Volume uplift vs baseline, by mechanic



Promotion ROI by mechanic and margin assumption (dashed = break-even)



Dip and cannibalisation (units vs baseline)

Category	Week after promo	Other products during promo
Bag Snacks	+1.0%	-0.1%
Cold Cereal	+4.5%	+0.2%
Frozen Pizza	+10.0%	+0.8%
Oral Hygiene Products	+0.1%	-0.2%

So what

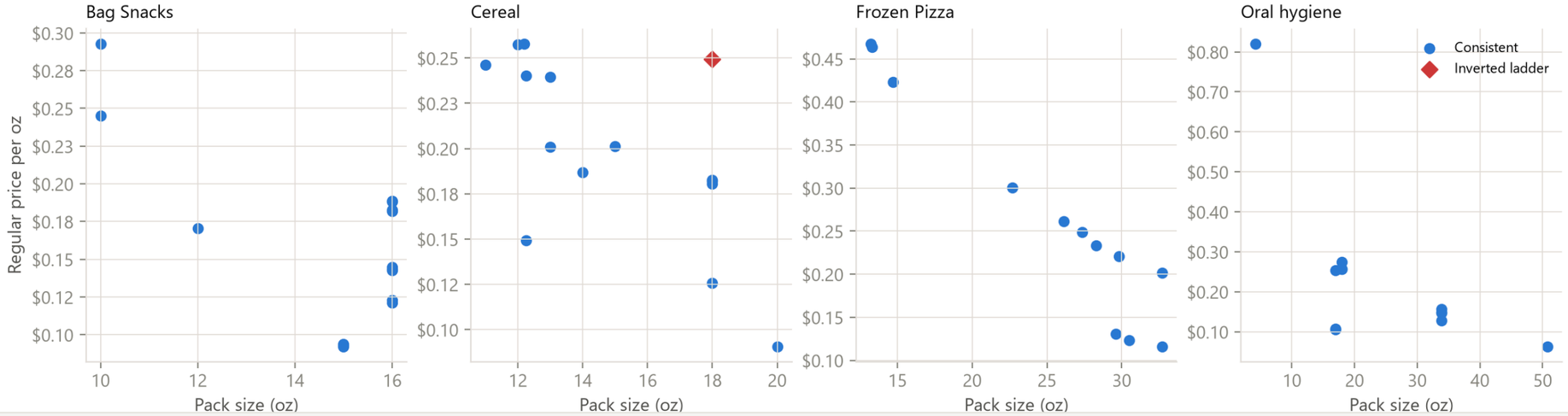
- A price cut alone lifts volume +33% but gives away more margin than it creates: ROI -60% at 30% margin.
- Overall, promotions return -29% on discount cost at 30% margin (-53% at 20%, -5% at 40%).
- No stock-up dip is visible: the week after a promotion sells +4% vs baseline; other products of the category stay at +0.2%, so cannibalisation is negligible.
- Feature and display fees are not in the data, so their ROI is optimistic.

Source: dunnhumby 'Breakfast at the Frat' · sql/02_promo_baseline.sql, src/promo.py · margin assumption: data/ASSUMPTIONS.md

1 pack breaks the price ladder; 6 products look under-priced for their low price sensitivity

A healthy ladder gives bigger packs a lower price per ounce. Prices are regular (base) prices so promotions do not distort the ladder; index 100 = sub-category average; liquids are converted to fluid ounces.

Regular price per ounce by pack size (red diamond = bigger pack costs more per oz than a smaller one of the same brand)



Flags and opportunities (BR-08, BR-11)

Category	Brand	Product	Price/oz	Index	Elasticity	Opportunity
Bag Snacks	Private Label	Pl Pretzel Sticks 15 oz	$0.092	70	-0.64	Price up candidate: low sensitivity, below-average price per oz
Bag Snacks	Private Label	Pl Twist Pretzels 15 oz	$0.094	71	-0.51	Price up candidate: low sensitivity, below-average price per oz
Bag Snacks	Private Label	Pl Mini Twist Pretzels 15 oz	$0.093	70	-0.68	Price up candidate: low sensitivity, below-average price per oz
Bag Snacks	Shultz	Shurgd Pretzel Sticks 16 oz	$0.123	92	-0.23	Price up candidate: low sensitivity, below-average price per oz
Bag Snacks	Shultz	Shurgd Mini Pretzels 16 oz	$0.121	91	-0.30	Price up candidate: low sensitivity, below-average price per oz
Cold Cereal	General Mills	Gm Cheerios 18 oz	$0.249	118	-1.92	Fix ladder: bigger pack costs more per oz
Cold Cereal	Kellogg	Kell Froot Loops 12.2 oz	$0.258	117	-3.04	Sensitive: hold price, use targeted promotions
Frozen Pizza	King	Nwmn Own Pepperoni Pizza 13.2 oz	$0.468	225	-3.00	Sensitive: hold price, use targeted promotions
Frozen Pizza	King	Nwmn Own 4 Cheese Pizza 13.3 oz	$0.464	223	-3.31	Sensitive: hold price, use targeted promotions

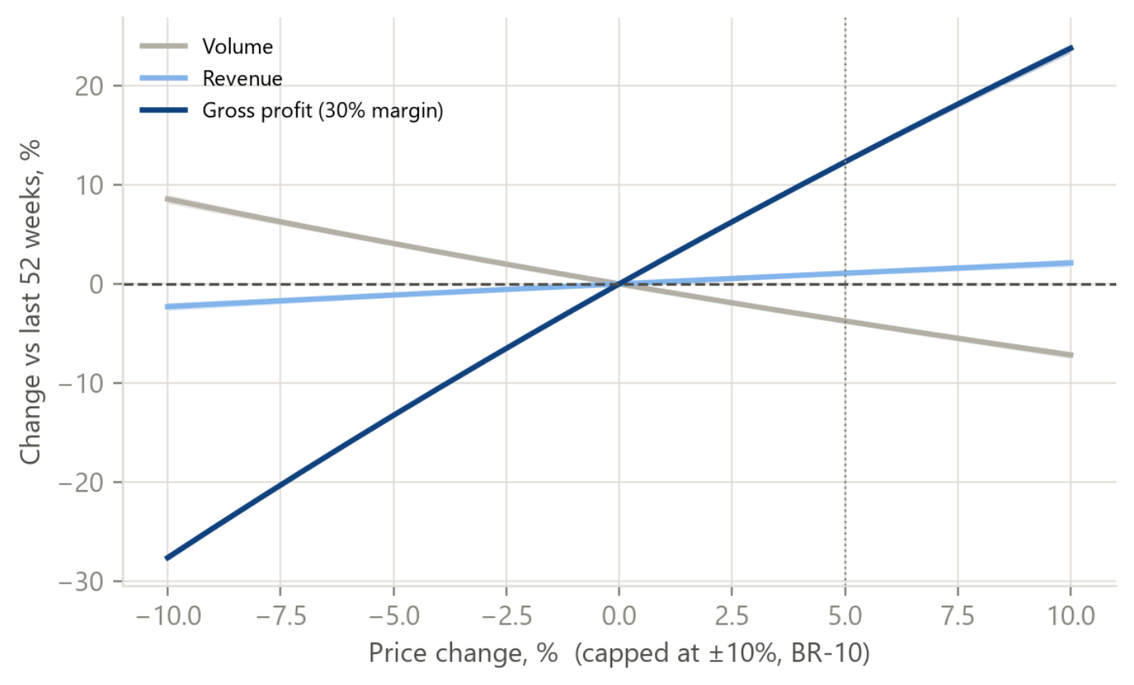
Source: dunnhumby 'Breakfast at the Frat' · sql/04_price_per_unit.sql, src/ppa.py

+5% on General Mills (cereal): volume -3.7%, revenue +1.1%, gross profit +12.3%

REAL DATA + MODEL

The largest published brand, elasticity -0.78 (95% CI -0.82 to -0.74). On last-52-week revenue of $1,704,883, that is +$18,399 revenue and +$62,962 gross profit per year at a 30% margin.

Cereal | General Mills: effect of a price change (bands = 95% interval)



+5% on every published brand (30% margin, range from the 95% interval)

Brand	Volume	Revenue	Gross profit
Bag Snacks | Shultz	-1.5% (-2.0% to -1.0%)	+3.4%	+14.9%
Bag Snacks | Private Label	-2.9% (-3.5% to -2.3%)	+1.9%	+13.3%
Cereal | General Mills	-3.7% (-3.9% to -3.6%)	+1.1%	+12.3%
Oral hygiene | Chattem	-4.0% (-4.9% to -3.1%)	+0.8%	+12.0%
Oral hygiene | Private Label	-4.9% (-5.2% to -4.6%)	-0.2%	+10.9%
Cereal | Private Label	-6.8% (-7.3% to -6.4%)	-2.2%	+8.7%
Bag Snacks | Mksl	-7.0% (-8.1% to -5.9%)	-2.4%	+8.5%
Bag Snacks | Frito Lay	-7.1% (-8.0% to -6.3%)	-2.5%	+8.3%
Oral hygiene | P&G	-8.3% (-8.7% to -7.9%)	-3.7%	+7.0%
Bag Snacks | Snyder's	-8.6% (-9.2% to -8.0%)	-4.0%	+6.7%
Cereal | Post Foods	-8.8% (-9.7% to -8.0%)	-4.3%	+6.4%
Cereal | Kellogg	-9.4% (-9.8% to -8.9%)	-4.9%	+5.7%
Oral hygiene | Warner	-9.4% (-10.0% to -8.9%)	-4.9%	+5.6%
Frozen Pizza | Private Label	-12.0% (-12.8% to -11.1%)	-7.6%	+2.7%
Frozen Pizza | Tombstone	-12.5% (-13.1% to -11.9%)	-8.1%	+2.1%
Frozen Pizza | Tony's	-12.7% (-13.8% to -11.6%)	-8.4%	+1.8%
Frozen Pizza | King	-13.7% (-15.0% to -12.4%)	-9.4%	+0.7%

How it works (BR-10, BR-11)

Volume = $(1+p)^{e}$; revenue = $(1+p)^{1+e}$; gross profit = $(1+p)^{e}$ x $(m+p)/m$. Revenue rises with price only if $|e| < 1$; gross profit rises with a small increase while $|e| < 1/m$ (3.3 at 30% margin). Only published elasticities, ±10% max, no competitor reaction: a first-order estimate for a proposal, not a forecast.

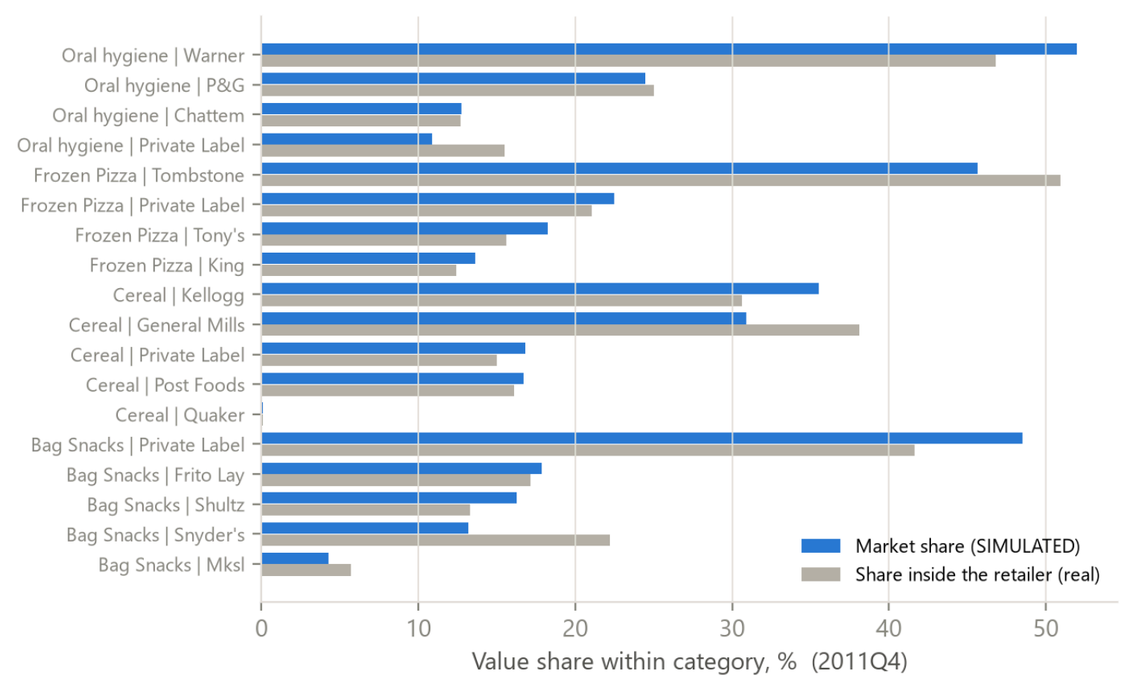
Source: dunnhumby 'Breakfast at the Frat' · src/simulator.py; DAX 'Sim *' measures in Power BI

Market view: how an external panel feed plugs into the product

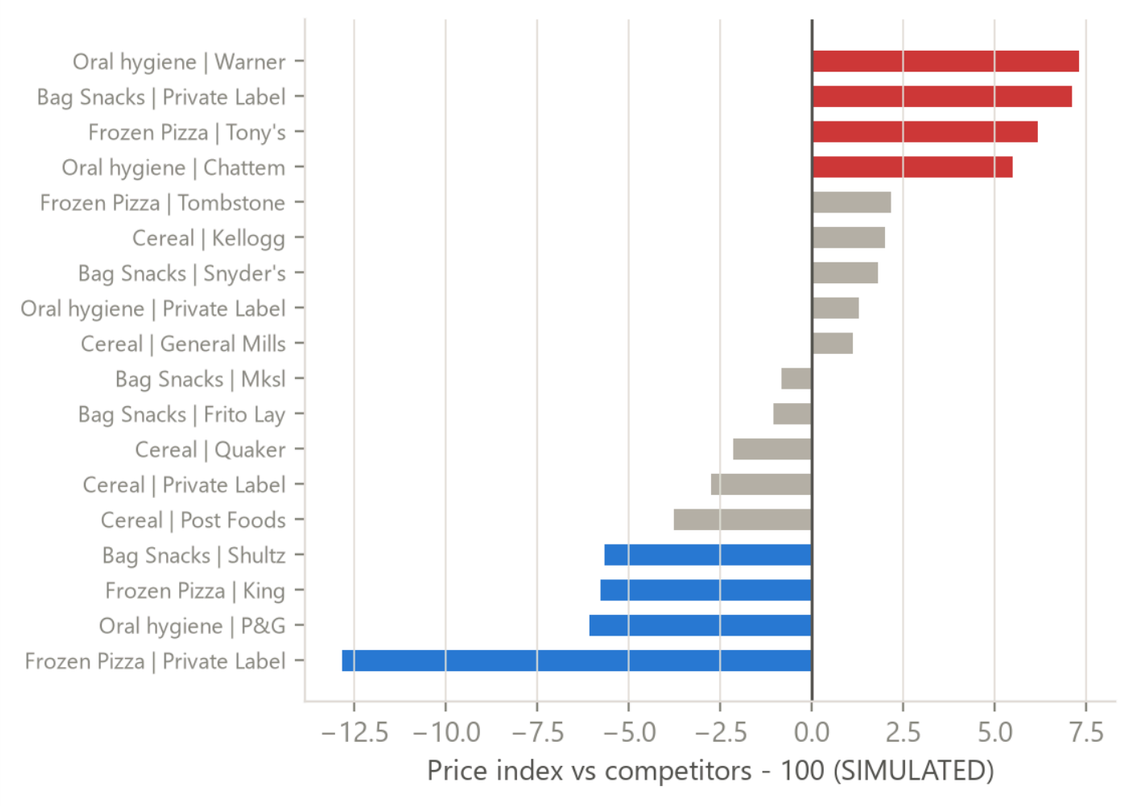
SIMULATED

SIMULATED data (seed 42), calibrated on the retailer's real brand shares. It shows the data model and the measures, not a market reading: do not quote these numbers.

Market share (SIMULATED) vs share inside the retailer (real)



Price index vs competitors (SIMULATED; red > 105, blue < 95)



Why simulate, and what changes with real panel data

NielsenIQ / Circana data is paid. The simulated feed has the same shape (brand x quarter: value, share, competitor price, numeric and weighted distribution), so the model, measures and pages are ready for a real feed. With real data, this page would answer: are we losing share where our price index is highest?

Same method in every market: Ohio shoppers are more price-sensitive than Texas in all 4 categories

REAL RETAIL DATA

Each US state in the data plays the role of a market. The pipeline, the business rules and the report are reused unchanged; only the market filter changes. This is how an RGM product scales across European markets. Indiana and Kentucky have fewer than 5 stores: not published (BR-06).

Category elasticity by market (state), with 95% intervals



Standard in every market

- Same SQL, Python and DAX; same business rules BR-01 to BR-13; same report pages.
- A market is a filter: no code change to add one (BR-12).
- Each market is published only on its own evidence: 8 of 16 market x category estimates pass.

What varies

- The size of the elasticity: largest gap between markets per category Bag Snacks 0.14, Cereal 0.19, Frozen Pizza 0.89, Oral hygiene 0.18.
- Sample size: small markets give wider intervals, so fewer publishable segments.
- In Europe: currency, VAT, retailer mix and promotion rules differ, and enter as market parameters.

Source: dunnhumby 'Breakfast at the Frat' · src/elasticity.py (level = Market)

From analysis to product: stories, rules and tests a development team can build on

The analysis is packaged as product documentation: user stories with priorities, testable acceptance criteria, numbered business rules and a test plan. Full documents in product/.

User stories (10)

- **US-01** As a Revenue Manager, I want each elasticity with its 95% range and sample size, so that I know where a price move is low-risk.
- **US-03** As a Trade Marketing Manager, I want uplift and ROI per mechanic and brand, so that I move spend to what pays back.
- **US-05** As a Revenue Manager, I want to simulate up to ±10% with a range, so that I can prepare a pricing proposal.
- **US-09** As a Product Owner, I want to add a market by changing a filter, so that the product scales at low cost.

Acceptance criteria (Given / When / Then)

AC-05.1 Given a published elasticity -1.5, +5% and a 30% margin, when the simulator runs, then volume = -7.06%, revenue = -2.41%, gross profit = +8.43%.

AC-05.4 Given two segments or a non-published one selected, when the page renders, then outputs are blank and the status explains why.

AC-03.2 Given base $4.00, promo $3.00, 100 units, baseline 40, margin 30%, then ROI = -28%.

Business rules (13)

- **BR-03** Baseline = average of clean non-promo weeks within ±8 weeks; ≥ 4 weeks or no baseline.
- **BR-06** Publish an elasticity only with ≥ 1,000 rows, ≥ 52 weeks, ≥ 5 stores, price variation ≥ 5%, $p < 0.05$, negative sign, interval ≤ 1.5. Positive sign = analyst review.
- **BR-10** Simulator: published segments only, ±10% max, ranges always shown.
- **BR-13** Every page tagged REAL / PUBLIC / SIMULATED.

Test plan

- 25 automated checks (`src/check_results.py`): rule unit tests, reconciliations, output assertions.
- DAX vs Python: each key measure compared with the Python value at release.
- Method test on simulated data: 9 of 10 known elasticities inside the 95% interval.
- 41 measures documented in `MEASURES_DICTIONARY.md`.

Methods, limits and how to reproduce

What the numbers can and cannot say, where every input comes from, and how to rerun everything from a clean clone.

Sources and licences

- **dunnhumby** *Breakfast at the Frat*, Source Files terms of use: raw data not redistributed; scripts and aggregates only.
- **Eurostat** HICP, CC BY 4.0.
- **Lingaro** public website, facts quoted with URL and date (2026-11-12).
- **Simulated** market data and test sample, seed 42, labelled everywhere.

Reproduce

`python src/run_all.py` (real data) or `python src/run_all.py --sample` (simulated sample, no download). Power BI: open `powerbi/LingaroRGM.pbip`, refresh.

Limits

- **Elasticity is observational**: a correlation on historical prices set by the retailer, not a controlled test. Controls reduce, but do not remove, confounding (unobserved competitor moves, local events).
- **Old US data** (2009-2012, one retailer): it demonstrates the method, not today's European elasticities.
- **Margin is assumed** (30%); feature and display fees are unknown, so their ROI is optimistic.
- **Market data is simulated**; no market conclusion is drawn from it.
- The simulator ignores competitor and retailer reactions and is capped at ±10%.

Methods

- SQL (DuckDB): baseline with window functions, hierarchy ROLLUP, price per oz, quality checks.
- Python: statsmodels OLS, log-log, store x product and month effects, trend, store-clustered errors.
- Data quality: rows flagged, not deleted (BR-07); duplicate store records and phasing-out products found and handled.
- Specification check: with separate store and product effects, bag snacks came out at -0.65, because Texas stores carry only 9 of the 15 snack products. One effect per store x product pair fixes this (-1.23, consistent with Ohio and Texas on their own).
- Validation: 25 automated checks pass; on simulated data with known elasticities, 9 of 10 true values fall inside the 95% interval.
- Power BI: star schema, 41 DAX measures, what-if parameters, TMDL + PBIR generated by script.

AI note

AI-assisted workflow: I used Claude, mainly to write scripts faster. I drove the project: I defined the questions, chose and validated the data sources, made the analytical decisions and checked every result.

Full details: README.md, data/SOURCES.md, data/ASSUMPTIONS.md